Rylie Horning

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Education

- **Virginia Tech** Blacksburg, Virginia
  Bachelors in Engineering: Electrical Engineering, Minor: Computer Science *August 2024 – May 2028*
  GPA 3.66/4.00
- **New River Community College** Dublin, Virginia
  Associates of Arts and Sciences: Computer Science *August 2021 – May 2024*
  Completed concurrently with high school

Skills

- **Electrical:** Circuit Analysis and Design, LTspice, Schematic Design, PCB Assembly/Soldering, Sensors
- **Programming:** Python, C/C++, Verilog/VHDL
- **Embedded/Tools:** Arduino, ESP32, Raspberry Pi, Oscilloscope, Multimeter, Power Supplies
- **Professional:** Technical Documentation (ISO 9001), Cross-Functional Teamwork

Experience

- **Moog Inc.** Christiansburg, Virginia
  Supply Chain Intern *July 2025 - Present*
  - Create work instructions using Microsoft Word and ensure documents meet the ISO 9001 standard.
  - Used SAP to support internal workflow and reporting tasks as needed.
- **Virginia Tech Electric Service** Blacksburg, Virginia
  Engineering Intern *October 2024 - July 2025*
  - Revised campus power distribution schematics and facility electrical maps using AutoCAD Electrical.
  - Gathered equipment data using a Trimble GPS device for mapping purposes.
  - Supported installation and maintenance of campus electrical infrastructure.
- **Motion Control Systems** Radford, Virginia
  Test Engineering Intern *January 2024 - September 2024*
  - Conducted testing on motor control PCBs across varying voltages to evaluate performance and thermal behavior.
  - Analyzed test data to evaluate system performance and verify design behavior
  - Assembled and built motor control PCBs from schematics, including soldering and hardware integration.

Projects

- **Home Audio System** — Analog Circuit Design, LTspice, Active Filters, Class-D Amplifier, Arduino, PWM
  - Designed and integrated a multi-stage audio system with analog filtering, Class-D amplification, and microcontroller interfacing.
  - Verified performance and behavior through LTspice simulation and hardware validation.
  - Implemented MOSFET switching stage driven by Arduino-generated carrier (>80 kHz) and designed a Butterworth RLC low-pass filter to meet headphone output specifications.
- **Power Supply Failure Detection System -** Analog design, test and validation
  - Designed and tested an analog voltage monitoring circuit to detect out-of-range conditions.
  - Simulated edge cases in LTspice and compared results with physical oscilloscope measurements.
  - Implemented comparator and BJT-based switching for fault indication.
- **Wi-Fi "On Air" LED Controller** — Embedded systems, testing and validation
  - Designed and built an ESP32-based embedded system for remote LED control over Wi-Fi.
  - Developed and tested firmware to ensure reliable relay operation and network communication.